

REVISION HISTORY

Revision No./Version No.	Date	Prepared by	Approved by	Details of Change
1.0	09-Aug-2024	Gayatri Tirodkar	Anant Sham Yende (CTO)	

INTRODUCTION

This document provides the standard operating procedures (SOP) for database design and modeling within the organization to ensure the development of efficient, scalable, and maintainable database systems.

PURPOSE

The purpose of this SOP is to establish a standardized approach for designing and modeling databases to meet organizational requirements and support business processes effectively.

SCOPE

This SOP applies to all database administrators, developers, and technical teams involved in the design and modeling of databases within the organization.

1. Understand Database Requirements :-

1.1 Identify the data to be stored and its usage based on the documents shared and explained in the sensitization meet and in FRS shared.

1.2 Determine performance, security requirements.

Performance :-For new Server Hardware and Resource allocation as per requirement to be done. Additionally as per project requirement resource allocation for required table space can be considered.

Apply Indexing and Query Optimization ,

Security :- Access Control , Data Encryption, Backup and Recovery , Authentication an Authorization.

2. Choose Database Engine :-

- 2.1 Choose database engine as per SOP on "How to choose a Database engine".
- 2.2 Discuss and confirm the appropriate DBMS based on the requirement of the project.
(e.g. MySQL , PostgreSQL , MongoDB)

3. Logical Design :-

Define Tables and Relationships :-

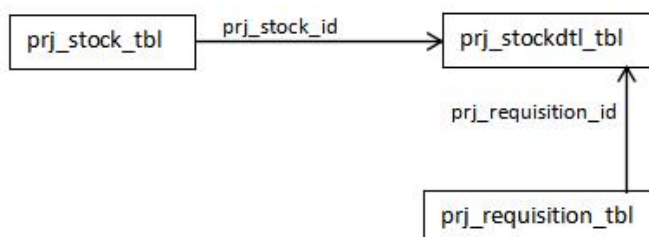
- 3.1 Convert entities to tables.
- 3.2 Prepare high level table link diagram based on the relationships of the table as per logical structure.
- 3.3 Once table link diagram is ready identify and list the columns. Maintain design date, author and link diagram as per revision in google sheet.
- 3.4 Define primary keys and foreign keys.
- 3.5 Establish table relationships and constraints.
- 3.6 Define entry/update comment on table.
- 3.7 Maintain the table with column and datatype and link diagram in google sheet
https://docs.google.com/spreadsheets/d/1ZZf2XK8GiRLdRjwHbF4nKeM7yipCA5Oi84MIsx_WH9c/edit?pli=1&gid=372233683#gid=372233683".
- 3.8 Maintain version of the tables in google sheet.
- 3.9 Review to see the data flow between tables is as per request and traceability is maintained.

Example :-

prj_stock_id = Stores SKU details with prj_stock_id

prj_requisition_tbl = Stores requisition details, to release stock from stores department.

prj_stockdtl_id = Stores available stock qty with Stock transactions against prj_requisition_id.



We can get details as below,

- i. SKU wise Stock :- From prj_stock_tbl considering prj_sku_id and prg_stock_id for that

SKU, Preset stock can be fetched from prj_stockdtl_tbl for the prj_stock_id from prj_stock_tbl.

ii. Utilization of SKU's against Requisition :- From prj_stockdtl_tbl against

prj_requisition_id we can get the stock utilized against requisition.

Also requisition details we can fetch from prj_requisition_tbl against prj_requisition_id from prj_stockdtl_tbl.

4. Database Design Approval :-

- 4.1 Take approval for table link diagram from CTO prepared on the logical structure.
- 4.2 Take database design approval from CTO to finalize on the DB structure.
- 4.3 In case , database migration is required then take approval from CTO with proper plan.

5. Physical Design :-

- 5.1 Create tables with relevant data types length and constraints for each attributes and maintain.
- 5.2 Design indexes to improve query performance.
- 5.3 Provide appropriate permissions to tables as per the user type. (Refer Database Coding Standards document in database users and user privilege section.)
- 5.4 Create and add master data and maintain the insert scripts.
- 5.5 Plan and execute data migration from existing systems if necessary.
- 5.6 Database Vertical Lead to review physical and logical design comparison with approved designs.

6. Review :-

- 6.1 Database Vertical Lead to review if physical design is in sync with the approved logical design.
- 6.2 Database Vertical Lead will ensure to get the dummy data filled in the tables and walk through the data to see traceability and ease of querying.

7. Continuous Improvement & Monitoring :-

Query cause of developments it may be observed that there are changes to processes or requests or missing design elements and/or additional provisions in data may be needed to ease the coding and faster verification retrieval. In such cases Database Vertical Lead should

STANDARD OPERATING PROCEDURE

DATABASE DESIGN AND MODELING

constantly monitor and take inputs from developers and accordingly get the changes incorporated with CTO Approval.